

1D Isentropic Nozzle Area-to-Mach Relation

i. Problem Statement

In the nozzles of air-breathing and rocket engines, flow speeds are high enough that compressibility effects are significant, and the Mach number M , the ratio of flow speed to the local speed of sound, governs how the flow behaves. Nozzle geometry and flow state are directly linked: for steady flow, mass conservation ties the cross-sectional area at each station to the Mach number there. This relationship is expressed through the area ratio A/A^* , where A^* is the area at which the flow reaches $M = 1$, which we assume is the throat of converging-diverging nozzle.

The Area-Mach relation gives A/A^* explicitly as a function of M , but the design question usually runs the other way: given a nozzle's geometry, what is the Mach number at each station? The relation cannot be inverted algebraically, and for every $A/A^* > 1$ it has two valid solutions, one subsonic and one supersonic. This document derives the relation and solves it numerically using the Newton-Raphson method.

The analysis assumes quasi-one-dimensional, steady, adiabatic, and isentropic flow of a calorically perfect gas. These assumptions break down in real world conditions, such as effects of viscosity, multi-dimensional flow, and/or shock formation within the nozzle. Hence, although nozzle flow isn't truly one dimensional, the quasi-1D isentropic model serves as a strong first layer of assessment for preliminary nozzle design.

ii. Area-to-Mach Relation

The derivation begins from the isentropic-stagnation relations for a calorically perfect gas, which follow from the steady, adiabatic energy conservation equation and the isentropic relation $p/\rho^\gamma = \text{const}$. The expression for stagnation density ratio and specific heat for constant pressure (c_p) is taken as shown and its derivations are presented in [1, Sec. 3.4]:

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{1}{\gamma - 1}}, \quad c_p = \frac{\gamma R}{\gamma - 1}$$

From mass conservation, $\rho_1 u_1 A_1 = \rho_2 u_2 A_2$. Applying this to nozzle throat and any general point in the nozzle, we get $\rho^* u^* A^* = \rho u A$. Note that $u^* = a^*$ at the sonic point. We can isolate for the area ratio:

$$\frac{A}{A^*} = \frac{\rho^*}{\rho} \cdot \frac{a^*}{u}, \quad \text{and note } \frac{\rho^*}{\rho} = \frac{\rho^*}{\rho_0} \cdot \frac{\rho_0}{\rho}$$

Where the stagnation density ρ_0 is constant throughout the field from the isentropic assumption. We have that

$$\frac{\rho_0}{\rho^*} = \left(1 + \frac{\gamma - 1}{2}\right)^{\frac{1}{\gamma - 1}} = \left(\frac{\gamma + 1}{2}\right)^{\frac{1}{\gamma - 1}} \implies \frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma + 1}\right)^{\frac{1}{\gamma - 1}}$$

and by substitution, we get

$$\frac{A}{A^*} = \left(\frac{2}{\gamma + 1} \right)^{\frac{1}{\gamma - 1}} \cdot \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{1}{\gamma - 1}} \cdot \left(\frac{a^*}{u} \right)$$

Hence, we require an expression for $\frac{a^*}{u}$. Since the flow is isentropic, it is adiabatic, and so the stagnation enthalpy is constant across the flow field:

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \implies c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2}$$

and using our expression for c_p we get

$$\frac{\gamma R}{\gamma - 1} T_1 + \frac{u_1^2}{2} = \frac{\gamma R}{\gamma - 1} T_2 + \frac{u_2^2}{2}.$$

From our calorically perfect gas assumption, we can apply the ideal gas equation of state, which lets us express the acoustic speed $a = \sqrt{\gamma R T}$ and so

$$\begin{aligned} \implies \frac{a_1^2}{\gamma - 1} + \frac{u_1^2}{2} &= \frac{a_2^2}{\gamma - 1} + \frac{u_2^2}{2} \\ \implies \frac{a^2}{\gamma - 1} + \frac{u^2}{2} &= \frac{a^{*2}}{\gamma - 1} + \frac{a^{*2}}{2} \\ \implies \frac{1}{u^2} \left(\frac{a^2}{\gamma - 1} + \frac{u^2}{2} \right) &= \frac{1}{u^2} \left(\frac{a^{*2}}{\gamma - 1} + \frac{a^{*2}}{2} \right) \\ \implies \frac{a^2}{u^2} \cdot \frac{1}{\gamma - 1} + \frac{1}{2} &= \left(\frac{a^{*2}}{u^2} \right) \cdot \left(\frac{1}{\gamma - 1} + \frac{1}{2} \right) \end{aligned}$$

Note that $\frac{a^2}{u^2} = M^{-2}$ and by multiplying both sides of the equation by 2:

$$\implies \frac{1}{M^2} \cdot \frac{2}{\gamma - 1} + 1 = \left(\frac{a^{*2}}{u^2} \right) \cdot \left(\frac{2}{\gamma - 1} + 1 \right)$$

By isolating the velocity ratio, we have

$$\begin{aligned} \left(\frac{a^*}{u} \right)^2 &= \frac{\frac{1}{M^2} \cdot \frac{2}{\gamma - 1} + 1}{\frac{2}{\gamma - 1} + 1} = \frac{\frac{1}{M^2} \cdot \frac{2}{\gamma - 1} + 1}{\frac{\gamma + 1}{\gamma - 1}} \\ &= \frac{1}{M^2} \cdot \frac{2}{\gamma + 1} + \frac{\gamma - 1}{\gamma + 1} \\ &= \frac{1}{M^2} \cdot \frac{2 + M^2(\gamma - 1)}{(\gamma + 1)} \\ &= \frac{1}{M^2} \cdot \frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \end{aligned}$$

We may now substitute this expression into our equation for the area ratio:

$$\begin{aligned}
\left(\frac{A}{A^*}\right)^2 &= \left(\frac{\rho^*}{\rho_0}\right)^2 \left(\frac{\rho_0}{\rho}\right)^2 \left(\frac{a^*}{u}\right)^2 \\
&= \left(\frac{2}{\gamma+1}\right)^{\frac{2}{\gamma-1}} \cdot \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{2}{\gamma-1}} \cdot \frac{1}{M^2} \left(\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2}M^2\right)\right) \\
&= \left(\frac{2}{\gamma+1}\right)^{\frac{2}{\gamma-1}+1} \cdot \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{2}{\gamma-1}+1} \cdot \frac{1}{M^2} \\
&= \frac{1}{M^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2}M^2\right) \right]^{\frac{2}{\gamma-1}+1}
\end{aligned}$$

Upon simplifying the exponent, we get our Area-to-Mach relation:

$$\boxed{\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2}M^2\right) \right]^{\frac{\gamma+1}{\gamma-1}}}$$

iii. Newton-Raphson Method

The area-to-Mach relation presents us with an explicit way of calculating an area ratio from a known Mach number, but the same cannot be said for calculating the Mach number provided a certain area ratio. For many nonlinear equations, including this one, a closed form inverse function does not exist, and so the Mach number must instead be found numerically. In this case, the Newton-Raphson method approximates a solution numerically by an iteration of linear approximations, and in this section we describe its setup. A proof of convergence is beyond the scope of this report; see [2]. In section (iv) we discuss its application to this particular nozzle flow problem.

First, suppose f is a function where r is its root, meaning $f(r) = 0$ and we are trying to approximate r . We can start with an initial estimate x_0 , and so we can write $r = x_0 + p_0$ where p_0 is the error between r and our initial guess. Using a linear tangent line approximation we can write

$$0 = f(r) = f(x_0 + p_0) \approx f(x_0) + p_0 \cdot f'(x_0) \implies p_0 \approx -\frac{f(x_0)}{f'(x_0)}$$

Now, we get that

$$r \approx x_0 - \frac{f(x_0)}{f'(x_0)} = x_1$$

where x_1 is our next guess for r , meaning $r = x_1 + p_1$ where p_1 is the error between r and our guess x_1 . Similarly, we can use a tangent line approximation to get our next guess x_2 , and so on. So, in general,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

This iteration is then stopped when the difference between a guess and the next is within some chosen tolerance. Meaning, when $|x_{i+1} - x_i| < \varepsilon$.

When the initial estimate is sufficiently close to a simple root, the method converges rapidly. Convergence is not guaranteed: for instance, if $f'(x_i)$ is near-zero, then $p_i \approx -f(x_i)/f'(x_i)$ becomes very large, sending x_{i+1} far from the root and possibly causing the method to diverge [2, Sec. 2.3].

iv. Solving for Mach Number

Before applying Newton-Raphson onto the area-to-mach relation, we must first observe that the area-to-Mach relation is not one-to-one. A/A^* reaches its minimum value of 1 at $M = 1$ and increases on either side, so every $A/A^* > 1$ corresponds to two Mach numbers: one subsonic and one supersonic. Physically, a converging-diverging nozzle carries subsonic flow upstream of the throat and, provided the pressure is low enough that no shock forms, supersonic flow downstream of it.

Since Newton-Raphson converges for one root per run, we require two runs, each with a different initial guess: one subsonic and one supersonic. First, define the residual function f_A such that

$$f_A(M) = \frac{1}{M^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{\gamma-1}} - \left(\frac{A}{A^*} \right)_{\text{target}}^2$$

Note, we call this function residual since it is the difference of the squared area ratio of the Mach estimate and the squared target ratio. Now, we take the derivative of this function with respect to M to get $f'_A(M)$:

$$f'_A(M) = \frac{d}{dM} \left[\frac{1}{M^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{\gamma-1}} \right]$$

Let $C = \frac{2}{\gamma+1}, \quad n = \frac{\gamma+1}{\gamma-1}$

By substitution,

$$\begin{aligned} f'_A(M) &= C^n \left[\frac{-2}{M^3} \left(1 + \frac{\gamma-1}{2} M^2 \right)^n + \frac{2nM}{M^2} \left(1 + \frac{\gamma-1}{2} M^2 \right)^{n-1} \left(\frac{\gamma-1}{2} \right) \right] \\ &= \frac{C^n}{M^3} \left(1 + \frac{\gamma-1}{2} M^2 \right)^{n-1} \cdot \left[-2 \left(1 + \frac{\gamma-1}{2} M^2 \right) + (\gamma+1)M^2 \right] \end{aligned}$$

Observe that the bracketed term can be simplified:

$$\begin{aligned} -2 \left(1 + \frac{\gamma-1}{2} M^2 \right) + (\gamma+1)M^2 &= -2 + (-\gamma+1)M^2 + (\gamma+1)M^2 \\ &= 2M^2 - 2 \\ &= 2(M^2 - 1) \end{aligned}$$

So, the first derivative of f_A is

$$f'_A(M) = \frac{2(M^2 - 1)}{M^3} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}} \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{2}{\gamma-1}}$$

Observe that as $M \rightarrow 1$, f'_A tends to zero which is where Newton-Raphson is least reliable.

Now, we have the functions necessary to iterate from one guess to the next in the context of the nozzle flow. However, even though the area-to-Mach relation has two physical solutions, the equation mathematically has four solutions since M appears as a squared quantity. Hence, there are two negative and two positive (Fig. 1)

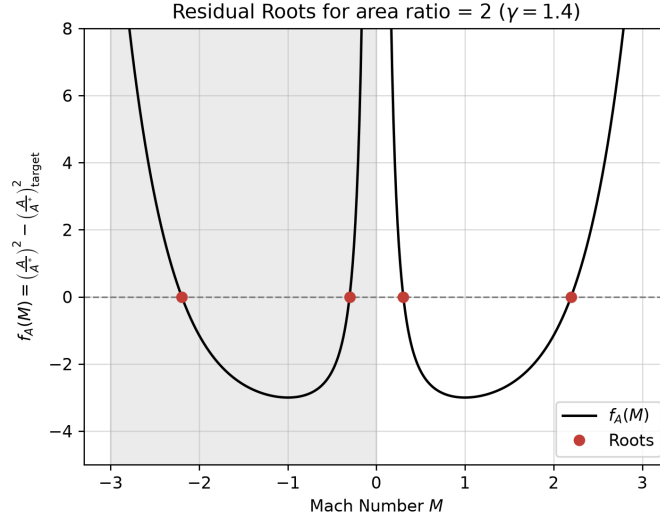


Figure 1: Plotting the residual function over a Mach number sweep reveals four real solutions. Only positive roots are physically plausible.

In fact, this problem was initially realized when the Newton Raphson solver converged to the negative roots of the residual function, an issue not originally realized since only physical solutions were considered. Since Newton-Raphson converges efficiently for guesses already close to the root, limiting behaviours of the area-to-Mach relation were used to find good initial guesses. This way, initial guesses already close to their respective roots would not converge to their negative counterparts.

First, for the subsonic initial guess, the limit behavior as $M \ll 1$ was

observed:

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}} \approx \frac{1}{M} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

Isolating for $M_{0,1}$, we have an expression for the *subsonic* initial guess:

$$M_{0,1} \approx C^{\frac{n}{2}} \cdot \left(\frac{A}{A^*} \right)^{-1}, \quad \text{where } C = \frac{2}{\gamma+1}, \quad n = \frac{\gamma+1}{\gamma-1}$$

Observe that $C^{n/2} < 1$ and since the area ratio is always above 1, we get $M_{0,1}$ is always subsonic as an initial guess.

Now, for the supersonic initial guess, the limit behavior as $M \gg 1$ was observed:

$$\begin{aligned} \frac{A}{A^*} &= \frac{1}{M} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}} \\ &\approx \frac{1}{M} \left(\frac{2}{\gamma+1} \cdot \frac{\gamma-1}{2} \cdot M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}} \\ &= \left(\frac{\gamma-1}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \cdot M^{\frac{2}{\gamma-1}} \end{aligned}$$

Isolating for $M_{0,2}$, we have an expression for the *supersonic* initial guess:

$$M_{0,2} \approx \left(n^{\frac{n}{2}} \cdot \frac{A}{A^*} \right)^{\frac{1}{n-1}}, \quad \text{where } n = \frac{\gamma+1}{\gamma-1}$$

Since $n > 1$ and the area ratio always above one, $M_{0,2}$ is always supersonic.

To summarize, when the Newton-Raphson solver is called to find the subsonic and supersonic solution for a target nozzle area ratio, it will first calculate initial guesses $M_{0,1}, M_{0,2}$ for the subsonic and supersonic iterations, respectively. Then, it will use the residual function f_A and its first derivative f'_A evaluated at the initial guesses to estimate the next guesses, which are then used to compute better approximations. This will be iterated until $|M_{i+1} - M_i| < \varepsilon$ where the tolerance ε in the solver code is set to 1e-8.

v. Solver Implementation

This section discusses the expected inputs and outputs of the python program.

Inputs. The implemented program asks the user to input a *nozzle area ratio* A/A^* and the known *specific heat ratio* γ of the gas.

Outputs. The Newton-Raphson solver is called to solve for the *subsonic and supersonic Mach number* roots M_1, M_2 . For each Mach number, the following *stagnation ratios* are calculated:

$$\begin{aligned}\frac{T_0}{T} &= \left(1 + \frac{\gamma - 1}{2} M^2\right) \\ \frac{p_0}{p} &= \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}} \\ \frac{\rho_0}{\rho} &= \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{1}{\gamma - 1}} \\ \frac{a_0}{a} &= \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{1}{2}}\end{aligned}$$

In order: Temperature, Pressure, Density, and Acoustic Speed ratio. A worked example is given in the following section.

vi. Results and Verification

First, a solved example with area ratio $A/A^* = 2$, $\gamma = 1.4$ is presented as a concrete example for the outputs described in section (v).

Table 1: Solved Mach number for area ratio $A/A^* = 2$ and calculated stagnation over static property ratios. See section (v) for more information regarding outputs.

	Mach	$\frac{T_0}{T}$	$\frac{p_0}{p}$	$\frac{\rho_0}{\rho}$	$\frac{a_0}{a}$
Subsonic	0.3059	1.0187	1.0671	1.0474	1.0093
Supersonic	2.1972	1.9655	10.6459	5.4163	1.4020

The solver is verified with three checks. First, a forward check evaluates the area-Mach relation from section (ii) at tabulated Mach numbers, confirming that the implementation reproduces the tabulated area ratios. Second, the analytic derivative $f'_A(M)$ from section (iv) is compared against a finite-difference approximation (`np.gradient`) over a sweep of Mach numbers. Because the derivative passes through zero at $M = 1$ and grows large as $M \rightarrow 0$, the error is reported as a max relative error in the max norm. Third, the Newton-Raphson solver is given tabulated area ratios as input, and each recovered Mach number is compared with the corresponding tabulated value. Since the table reports area ratios to four decimal places, each case is judged against the error that this rounding alone would produce, which grows sharply near $M = 1$.

Reference values are taken from Table B.1 of *Gas Dynamics* by John and Keith [1], which tabulates isentropic flow properties for $\gamma = 1.4$. Although the derivation holds for general γ , **all three tests use $\gamma = 1.4$** .

Forward Relation Test. This test confirms that the implemented relation reproduces tabulated area ratio values from tabulated Mach numbers by direct computation. Eight tabulated Mach numbers (subsonic, near-sonic, and supersonic) were used in the relation described in section (ii) to solve for nozzle area ratio, and then compared to the tabulated area ratios in Table B.1 [1]. A tolerance of 5×10^{-5} was used which is the most the table’s rounding can account for as it reports area ratios to four decimal places. The maximum error was 4.38×10^{-5} which was within the tolerance. The errors at $M = 2$ and $M = 5$ are effectively zero because the exact area ratios there, 1.6875 and 25, are represented in the table without rounding.

Derivative Test. This test was used to verify the analytical derivative expression derived in section (iv). A sweep of Mach numbers from 0.01 to 5 with a step size of 1×10^{-6} was used to compute the squared area ratio and its derivative at each step using the analytical expression, and the resulting derivative array was compared to `np.gradient` applied on the computed squared area ratio array. An edge order of 2 was used since one-sided differences at the endpoints are otherwise first-order accurate. A tolerance of 1×10^{-6} was used. The error was reported as relative error in the max norm since the derivative expression grows large when M is small and is zero when $M = 1$. With a sweep of 4,990,001 points, the max relative error was 4.00×10^{-8} , well within the tolerance. Reducing the step size by a factor of 10 resulted in a max relative error of 3.77×10^{-10} , which is two orders of magnitudes lower than the first trial, indicating a second order correspondence.

Inverse Newton–Raphson Test. This test confirms that the solver recovers the correct Mach number on each branch. The tabulated area ratios from Table B.1 [1] were used as solver inputs, and each recovered Mach number was compared with the tabulated value on the matching branch: subsonic for tabulated $M < 1$ and supersonic otherwise.

A single fixed tolerance is not appropriate for this comparison. Because the table reports area ratios to four decimal places, each input carries a rounding error of up to 5×10^{-5} , and the resulting error in the recovered Mach number depends on the local slope of the area–Mach relation:

$$\delta M \approx \frac{5 \times 10^{-5}}{|d(A/A^*)/dM|}, \quad \frac{d(A/A^*)}{dM} = \frac{f'_A(M)}{2\sqrt{(A/A^*)^2}}.$$

Near $M = 1$ the slope approaches zero, so even exact computation can differ from the table by a comparatively large amount. Each case was therefore judged against its own expected rounding error. Table 2 summarizes the results.

Table 2: Inverse check: Newton–Raphson results compared with tabulated values [1, Table B.1], $\gamma = 1.4$.

Table M	Input A/A^*	Solver M	Error	Expected error	Iterations
0.20	2.9635	0.2000	1.42×10^{-6}	3.54×10^{-6}	4
0.50	1.3398	0.5000	2.29×10^{-5}	2.61×10^{-5}	5
0.80	1.0382	0.8001	7.24×10^{-5}	1.21×10^{-4}	6
0.96	1.0014	0.9595	5.10×10^{-4}	7.24×10^{-4}	9
1.50	1.1762	1.5000	4.87×10^{-5}	7.40×10^{-5}	8
2.00	1.6875	2.0000	0	3.56×10^{-5}	8
3.00	4.2346	3.0000	7.96×10^{-6}	1.24×10^{-5}	7
5.00	25.000	5.0000	8.88×10^{-16}	2.50×10^{-6}	6

All eight cases pass within their expected rounding error. The near-sonic case, $M = 0.96$, has the largest error, roughly an order of magnitude above the others, and also required the most iterations. Both effects share the same cause: near $M = 1$ the area ratio changes very little with Mach number (Fig. 2), which amplifies the table’s rounding error and reduces the effectiveness of each Newton step, as discussed in Section iv. The case at $M = 2$ shows zero error because its exact area ratio, 1.6875, is represented in the table without rounding.

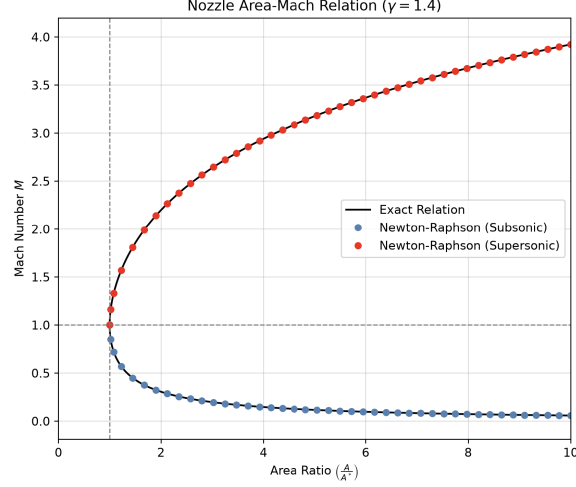


Figure 2: Area to Mach relation plotted showcasing the exact relation and the Newton-Raphson solved Mach numbers on top. Note the curve becomes nearly vertical $A/A^* \rightarrow 1$.

vii. Summary

This report derived the quasi-one-dimensional isentropic area-Mach relation and solved its inverse numerically with the Newton-Raphson method. Because the relation depends on M only through M^2 , the residual function f_A has four real roots, of which only the two positive ones are physical, and its derivative vanishes at $M = 1$, where Newton’s method is least reliable. Initial guesses derived from the relation’s small M and large M limits address both issues: each guess is guaranteed to start on its intended regime close to the actual root.

Three verification checks confirmed the implementation against Table B.1 of [1]. The forward relation reproduced tabulated area ratios to within 4.38×10^{-5} , inside the table’s own rounding precision. The analytic derivative agreed with a finite-difference approximation to a relative error of 4.0×10^{-8} , with the error decreasing at a second order rate. The solver recovered the tabulated Mach number in all eight cases within the error attributable to table rounding, with the largest deviation at the near-sonic case, as predicted by the flat slope of the relation near $M = 1$.

These results verify the solver for the quasi-one-dimensional isentropic model but they do not test that model against real nozzle flows, which depart from it through viscous effects, multidimensional flow, and shocks as discussed in section (i). A natural next-step is to apply the solver along a specified nozzle contour $A(x)$ to compute the Mach number, pressure, and temperature distributions through a complete nozzle.

References

- [1] J. E. A. John and T. G. Keith, *Gas Dynamics*, 3rd ed. Upper Saddle River, NJ: Pearson Prentice Hall, 2006.
- [2] “The Newton–Raphson Method,” course notes, MATH 104, University of British Columbia. [Online]. Available: <https://personal.math.ubc.ca/~anstee/math104/newtonmethod.pdf>. Accessed: Oct. 1, 2026.